Aniket Maity

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.26
DAV Model School IIT Kharagpur	Class XII (CBSE)	2023	94.16%
DAV Model School IIT Kharagpur	Class X (CBSE)	2021	95.83%

SCHOLASTIC ACHIEVEMENTS (BEFORE UNDERGRADUATE)

- **KVPY Qualified 2021:** Qualified in Kishore Vaigyanik Protsahan Yojana (KVPY) with an All India Rank 1688.
- **JBNSTS Scholarship 2021:** Awarded Jagadis Bose National Science Talent Search junior scholarship (JBNSTS, https://www.jbnsts.ac.in).
- **NGSE (Advanced) Qualified 2015:** Certificate of Excellence from National Genius Search Foundation (NGSF, https://www.ngsfindia.org) for qualifying in National Genius Search Examination [Advanced].
- **VVM Certificate 2020:** Certificate of Merit in Vidyarthi Vigyan Manthan (VVM) for securing district rank 3 (https://www.vvm.org.in).
- **SAT Top Performer 2023:** Recognized as SAT India Top Performer with a score of 1430/1600 with 98th Nationally Representative Sample Percentile and 780/800 Math score.

RELEVANT COURSES FROM UNDERGRADUATE CURRICULUM

- **Materials Science and Engineering (Theoretical Courses):**
- **Functional Materials Related Courses:** Physics of Solids, Semiconductor Materials, Soft and Porous Materials, Vacuum Technology in Semiconductors, Device Fabrication and Processing, Electronic Structure and Atomistic Modeling of Materials, Energy Materials and Devices, Polymer Science and Engineering.
- **Material Characterisation Related Courses:** Imaging in Materials Science, Diffraction and Spectroscopy, Theory and Practice of Atom Probe Tomography (ongoing).
- **Metallurgy Related Courses:** Iron Making, Steel Making, Mechanical Behaviour of Materials, Phase Transformations, Thermomechanical Processing, Fundamentals of Lightweight Alloys (ongoing), Imperfections in Crystalline Solids (ongoing).
- **Laboratory Courses:** Physics, Chemistry, Microstructure and Properties, Materials Extraction Synthesis and Characterisation, Mechanical Behavior, Deformation Processing and Heat Treatment, Functional Device Fabrication and Electrochemical Characterization (ongoing).
- **Other Relevant Courses:** Maths for Physics, Introduction to AI, Digital Fabrication, Modern Physics, Sustainability under a Changing Climate.

COURSES FROM ELECTRICAL ENGINEERING (MINOR)

- **Basic Courses:** Basic Electrical Engineering, Digital Circuits.
- **Advanced Courses:** Semiconductor Device Fundamentals (ongoing).

ADDITIONAL COURSES BEYOND CURRICULUM

Following courses are completed from Coursera:

- **Introduction to Machine Learning:** Authorized by Duke University, Course instructors - Prof. Genevieve M. Lipp, Prof. David Carlson, and Prof. Lawrence Carin.

Verification link: Completion Certificate for Introduction to Machine Learning.

- **Rust Fundamentals:** Authorized by Duke University, Course instructors - Prof. Alfredo Deza and Prof. Noah Gift.

Verification link: Completion Certificate for Rust Fundamentals.

UNDERGRADUATE RESEARCH PROJECT

BTech Research Project on **Single Crystal Metal-Organic Frameworks** during December 2025 to April 2026 under the guidance of Dr. Deepu J. Babu, Department of Materials Science and Metallurgical Engineering (MSME), IIT Hyderabad.

- *Outcome & Recognition:* Awarded the highest grade (**'A', 10/10**) for the contribution made and learning.
- *Key Contributions:* Synthesised millimetre sized crystals of ZIF-8. I am further expanding the study by performing gas permeation studies on the synthesised single crystals. This study will be useful for specialised gas-separation applications and in sensitive sensors.

INTERNSHIPS

Following internships are completed during undergraduate studies:

- **Fundamentals of Sputter Processing and Understanding Structure Zone Models**, under the supervision of Prof. Ayan Roy Chaudhuri, Materials Science Centre, IIT Kharagpur in December 2023.

- **All Solid-State Polymer Electrolyte**, under the supervision of Prof. Susanta Banerjee, Materials Science Centre, IIT Kharagpur from May 2024 to July 2024 (*published in a peer reviewed journal paper, details shown later*).

- **Optimisation of MFD Based Water Purification Process,** under the supervision of Prof. Praveen C. Ramamurthy, Department of Materials Engineering, Indian Institute of Science, Bengaluru from May 2025 to July 2025.

- **Flow Rate and Material Capacity Tuning for Microfluidic Water Purification,** under the supervision of Prof. Praveen C. Ramamurthy, Department of Materials Engineering, Indian Institute of Science, Bengaluru from May 2026 to July 2026.

PUBLICATIONS

Peer Reviewed Journals:

- Kamble R., **A. Maity**, S. Agarwal, D. Pradhan and S. Banerjee (2026), Tailoring Proton Transport in s-PVDF-HFP Membranes Through MOF Integration: A Route to Nafion-Free Fuel Cell Electrolytes, *Particle & Particle Systems Characterization*, Wiley, 43:e00238, https://doi.org/10.1002/ppsc.202500238.

(My contributions in this study are synthesising Fe-MIL-88B, fabricating composite membranes, and characterizing the membranes through Thermogravimetric Analysis (TGA), Differential Scanning Calorimetry (DSC) and Fourier Transform Infrared Spectroscopy (FTIR))

TECHNICAL SKILLS

- **Programming & Frameworks:** Python, MATLAB, Rust, TensorFlow, MLP, CNN.

- **Core Computational Skills:** Machine Learning, Numerical Modeling, Data Analysis.

- **Experimental & Lab Techniques:** Scanning Electron Microscopy (SEM), Powder X-Ray Diffraction (PXRD), Atom Probe, Sputtering Chamber, TGA-DSC, FTIR, Optical Microscopy.

POSITIONS OF RESPONSIBILITY

Events and Membership Coordinator: The Indian Institute of Metals (IIM) Student chapter at IIT Hyderabad.

EXTRACURRICULAR ACTIVITY

National Service Scheme (NSS), IIT Hyderabad